

RISK SCORING REPORT

Project FORESIGHT — Stage 8: Forecast & Inventory Risk Decisioning
Client: NorthBay Living | Zidio Development Internship | August 2026

1. Purpose

This report converts the chosen weekly demand forecast (LightGBM) and current inventory positions into stockout and overstock risk flags, recommended actions, and approximate rupee impact. It implements the decisioning grid required by the Project FORESIGHT brief.

2. Inputs Used

- Forecast: LightGBM model (test WAPE 19.96%, beating seasonal-naive 31.26%)
- Inventory: on_hand + on_order at the latest week
- Lead time: used to estimate demand during replenishment window
- Price: average unit price for ■ impact estimates

3. Risk Logic (Decision Grid)

Inventory position = on_hand + on_order. Demand during lead time = weekly forecast × (lead_time_days / 7). Weeks of cover = inventory position / weekly forecast.

Condition	Action
Stockout risk (inventory < demand during lead time)	REORDER
Overstock risk (weeks of cover > 12)	MARKDOWN / REDUCE
Neither risk	HEALTHY
Both flags (rare)	WATCH

Sales at risk ≈ forecast × avg_price for REORDER SKUs. Locked capital ≈ excess units (above 8 weeks cover) × avg_price × 0.6 (cost proxy).

4. Results — Latest Week Snapshot

Metric	Value
HEALTHY SKUs	193
REORDER SKUs	5
MARKDOWN / REDUCE SKUs	2
SKUs with Stockout Risk	5
SKUs with Overstock Risk	2
Total Sales at Risk	■ 99,745
Total Locked Capital	■ 197,699

5. Priority REORDER List (Highest Sales at Risk)

SKU	Category	Forecast	Inv. Pos.	Wks Cover	Sales at Risk (■)
SKU0144	Furniture	53.1	70	1.3	46,078
SKU0143	Small Appliances	113.0	239	2.1	42,261
SKU0199	Decor	81.4	65	0.8	5,990
SKU0073	Small Appliances	36.5	109	3.0	2,932
SKU0048	Small Appliances	60.4	172	2.8	2,483

6. Priority MARKDOWN / REDUCE List (Highest Locked Capital)

SKU	Category	Forecast	Inv. Pos.	Wks Cover	Locked Capital (₹)
SKU0190	Furniture	55.0	686	12.5	118,611
SKU0074	Furniture	8.1	120	14.9	17,205

7. Business Interpretation

- The majority of the assortment (193 of 200 SKUs) is currently in a healthy inventory position.
- Only 5 SKUs require urgent reorder attention. The top two (SKU0144 and SKU0143) account for most of the ₹99.7k sales-at-risk.
- Only 2 SKUs are clearly overstocked. SKU0190 alone holds ~₹1.19 lakh of estimated locked capital.
- Overall risk is concentrated in a small number of SKUs — this is actionable for the operations team.

8. Deliverable File

- sku_risk_scores.csv — full SKU-level risk table with forecast, inventory, flags, action, and ₹ impact

9. Next Steps (Stage 9)

- Build a simple Streamlit dashboard for non-technical users
- Prepare the executive readout summarising baseline vs model, WAPE, and risk findings
- Package all reports and notebooks for final submission

10. Conclusion

Stage 8 is complete. The LightGBM forecast has been combined with current inventory to produce clear REORDER / MARKDOWN / HEALTHY recommendations and approximate rupee impact. Risk is limited to 7 SKUs out of 200, giving the client a focused action list.